

Disproof of Conjecture 2617: a nonzero incidence-algebra radical

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Claim and counterexample. Conjecture 2617 asserts that the Jacobson radical of an incidence algebra of a locally finite poset is zero. Let $X = \{0 < 1\}$ and let the coefficient field be $\mathbb{F}_2$. The poset X is finite, hence locally finite. Its incidence algebra is

$$R = I(X, \mathbb{F}_2) = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in \mathbb{F}_2 \right\}.$$

Addition is entrywise and multiplication is the incidence convolution, which here is ordinary upper triangular matrix multiplication. Write

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0.$$

We prove $e \in J(R)$ using the definition of the Jacobson radical as the intersection of all maximal proper left ideals.

Proof of radical membership. Let M be an arbitrary maximal proper left ideal of R , and suppose $e \notin M$. The set

$$L = M + Re = \{m + re : m \in M, r \in R\}$$

is a left ideal containing M and containing e . Hence it strictly contains M , and maximality implies $L = R$. In particular there are $m \in M$ and $r \in R$ with $1 = m + re$. For every upper triangular matrix r , the product re is strictly upper triangular, so $(re)^2 = 0$. Because the characteristic is two, $m = 1 + re$ and

$$(1 + re)m = (1 + re)^2 = 1 + 2re + (re)^2 = 1.$$

The left-ideal property then gives $1 \in M$, contradicting properness. Thus $e \in M$ for every maximal proper left ideal M , so $e \in J(R)$. Since $e \neq 0$, the Jacobson radical is nonzero. This disproves the asserted universal vanishing, and therefore the conjecture.

Lean verification and semantic correspondence. The self-contained file `lean/Main.lean` imports only `Std`. It encodes the matrix with entries (a, b, c) by the three-bit integer $4a + 2b + c \in \{0, \dots, 7\}$. Boolean exclusive-or and conjunction implement the two operations of $\mathbb{F}_2$. The theorem `ring_laws` checks all unital-ring laws for these operations; additive inverse is the identity in characteristic two. Theorems `entries_supported`, `addition_is_pointwise`, and `multiplication_is_convolution` identify the ring operations with those of the incidence algebra. Theorems `every_incidence_function` and `entries_injective` verify that this encoding represents exactly all incidence functions, without missing or duplicating any.

The structure `LeftIdeal` quantifies over arbitrary predicates on the ring satisfying the actual ideal axioms. `MaximalProper` means proper and maximal among proper left ideals; `InJacobsonRadical` is their intersection. The definition `enlarge` constructs $M + Re$, including its closure proofs. The theorem `radical_contains_e` proves the argument above for every maximal proper left ideal. The final theorem `not_zero_radical_claim` proves the negation of radical vanishing in this specific incidence algebra. In particular, the formal argument does not infer Jacobson-radical membership merely from the equation $e^2 = 0$.

Reproduction and trust. Run `lean Main.lean` using Lean 4.19.0. It exits successfully. The principal theorem reports dependencies only on the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no additional axioms, unproved placeholders, or uses of `native_decide`. Finite algebraic identities are verified by kernel-checked `decide`; membership in all maximal left ideals is proved symbolically.